FRAMES n=4: two-phase precision polishing

Adaptive calibration follow-up, motivated by the completed 12-arm grid—not part of that original frozen grid

Calibration: 0/2 joint-gate passes. No eligible recovery winner; confirmation skipped.

One new arm, declared before its runs: gamma(t)=(4/(t+4))^0.75 for t<48,000, then gamma(t)=4/(t+4) for t>=48,000. The iteration clock and momentum state continue; no restart at the switch, no post-hoc rescaling and no TP projection.

Unchanged: alpha0=5e-5/64; alpha(t)=alpha0(t+1)^0.25; rho(t)=4/(t+8)^(2/3); batch=32; tau=4. Each trajectory starts afresh for 400,000 updates (12.8 million sampled rows), including its repeated 48k prefix.

Schedule	Channel	RMSE	TP norm.	Relative RMSE	Infidelity (1−F)	Joint gate
Prior best, single phase	0	3.09e-06	7.25e-05	7.31e-05	8.23e-08	FAIL
Prior best, single phase	1	2.79e-06	6.63e-05	6.59e-05	5.96e-08	FAIL
New, two phase	0	1.69e-06	5.04e-05	3.99e-05	6.43e-09	FAIL
New, two phase	1	1.80e-06	5.37e-05	4.23e-05	3.02e-09	FAIL

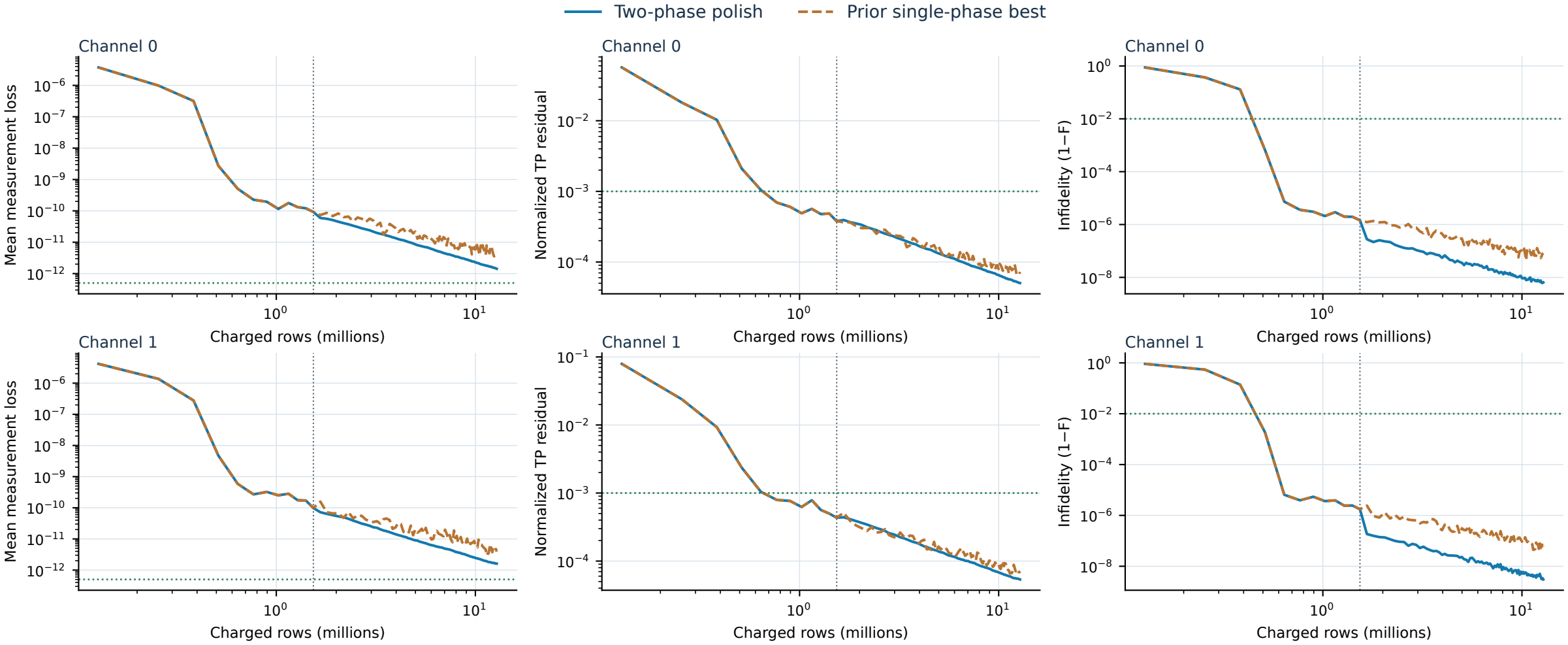
48k prefix: channel	Checkpoints compared	Physical metrics	Applied schedules	Max |RMSE difference|
0	13	MATCH	MATCH	0.00e+00
1	13	MATCH	MATCH	0.00e+00

Unchanged joint gate: full-design RMSE <=1e-6, normalized TP residual <=1e-3, process fidelity >=0.99. Relative RMSE = $||A(UU^H)-y||_2/||y||_2$. Gate decisions use full precision, not rounded table values.

Motivation: the prior best had very high fidelity and passed TP, but missed the RMSE threshold. Its small trace deficit and late fluctuations suggested testing smaller steps. This is an empirical hypothesis, not an established causal explanation or convergence theorem.

Paired trajectories at identical sampled-row budgets

Same physical data, initializer and declared sampling seed within each calibration channel; only gamma changes after 48k



Vertical dotted line: fixed switch after 48,000 updates (1.536 million rows). Horizontal dotted lines: physical gates. Mean measurement loss excludes the TP penalty. Penalized objectives are not used for ranking or comparison.

Zero-row initial points are omitted on logarithmic axes; plotted values are floored at 1e-16. Recorded checkpoints do not certify every intermediate iterate. Row counts exclude TP, LMO and metric work.

Confirmation, accounting and remaining uncertainty

No new tuning from confirmation; final held-out channels 1000–1019 remain unopened

Calibration: 0/2 joint-gate passes. No eligible recovery winner; confirmation skipped.

CONFIRMATION EXPLICITLY SKIPPED: this arm did not pass all three endpoint gates on both calibration channels. No recovery winner is declared; channels 2/3 were not opened to rescue this calibration. The completed negative result and the original 12-arm report are preserved.

Stage	Trials	Complete	Num. failures	Gate passes	Recorded rows (M)	Recorded updates
calibration	2	2	0	0	25.600	800,000
confirmation	SKIPPED	—	—	—	0	0
TOTAL NEW	2	2	0	0	25.600	800,000

All new starts and repeated prefixes are charged. The prior matched control used 25.6 million rows in the earlier round; reusing its archive adds zero optimizer work here. This polish is additional tuning cost, not a replacement for the original search budget.

The original twelve-arm grid was completed before this separate, adaptively motivated test. Only one new schedule was evaluated here. Success would support this frozen recipe on the tested cases, not an exhaustive optimum, a proof of nonlinear factor-method convergence, or a noise/higher-rank claim.

This remains a noiseless rank-one n=4 experiment (d=16; 8,192 physical Born probabilities; tau=4). Sampling seeds and unchanged native source are pinned; realized batches are not archived. Published Quiroga remains unresolved: no completed best-for-best or runtime-superiority claim.

Validation: 98 passed in 4.20s. Source 4ce1108a0692. Input/output SHA-256 hashes accompany this PDF.